

CHANDRAMITA BHATTACHARYA

AI ENGINEER | MULTIMODAL GENAI & COMPUTER VISION

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SUMMARY

AI Engineer with an M.Sc. in Cognitive Systems and hands-on experience turning visual data into practical AI applications. At Liebherr, I improved camera-based pedestrian detection, evaluated computer vision methods, and built LLM-powered agent workflows that analyse technical images and documents, retrieve relevant information, and respond to user requests through connected tools. I also bring experience in multimodal AI, human evaluation, and reliable software delivery, well suited to developing and assessing GenAI prototypes for automated visual content and future in-vehicle experiences.

TECHNICAL SKILLS

Programming & Media Processing: Python, NumPy, pandas, scikit-image

Multimodal GenAI & Agents: LangChain, LangGraph, Deep Agents, MCP/FastMCP

Computer Vision & XAI: OpenCV, YOLO/YOLOX, Segment Anything (SAM), PaddleOCR, InceptionV3, LIME

Cloud & Engineering Delivery: Azure, Databricks, Docker, Git, CI/CD, Agile

WORK EXPERIENCE

Data & AI Engineer Intern | Liebherr Digital Center, Ulm

Jul 2025 - Present

- Advanced the LiXplore camera system pedestrian-detection module by correcting fisheye-camera distortion and integrating a state-of-the-art YOLOX detector, supporting an industry-standard vehicle safety system.
- Conducted comparative evaluations of state-of-the-art OCR, object-detection and image-segmentation methods and curated specialised datasets to guide model selection and continuous algorithm improvement.
- Developed LLM-based document-intelligence prototypes for extracting layouts and tables from technical drawings and scanned documents; curated a table-parsing benchmark that identified preprocessing improvements and measurably enhanced retrieval behaviour and overall RAG accuracy.
- Built source-grounded knowledge-extraction workflows with Deep Agents, leveraging LangChain components and the LangGraph runtime, alongside MCP tools, LiteLLM routing and Langfuse tracing.

Associate - Technology Consulting | PricewaterhouseCoopers (PwC), India

Jul 2019 - Aug 2022

- Delivered and tested production web applications for international insurance and e-commerce clients using Angular, TypeScript and JavaScript, translating requirements into reliable releases.
- Instrumented an e-commerce platform with Google Analytics and Google Tag Manager event tracking and marketing automation to analyse user behaviour and support product decisions.

SELECTED PROJECTS

Multimodal Content Evaluation – Text-to-3D Generative Models | M.Sc. Thesis (1.0)

- Designed an LLM-driven benchmark for six generative models, generating 90 shape/colour/style prompts and evaluating multi-view renders with four metrics across semantic alignment, plausibility and mesh quality; produced a reproducible dataset/codebase and a manuscript submitted to WACV 2026.

Explainable AI – Image Segmentation for Visual Explanations | OpenCV, YOLO, SAM, LIME, PyTorch

- Benchmarked five segmentation approaches on 100 images, comparing explanation quality and runtime to assess their suitability for interpretable image classification.

Human Evaluation of AI – Generated Visual Content | Python, Gradio, pandas

- Co-created a dataset of 260 AI-generated portraits and conducted a three-phase annotation study, achieving 77.5% category agreement and identifying the hardest visual defects to classify.

Trust in Autonomous-Vehicle Interfaces | Unity, LimeSurvey, Python, Statistical Analysis

- Designed and evaluated Unity-based autonomous-vehicle interaction scenarios through a 46-participant user study, using statistical analysis to measure how anthropomorphic eHMIs influence passenger trust and user experience.

EDUCATION AND ADDITIONAL INFORMATION

- M.Sc. Cognitive Systems, Ulm University, Germany | 2022-2026:** Grade 1.5 (Thesis 1.0)
- B.Tech Computer Science, Institute of Engineering and Management, Kolkata | 2015-2019:** Grade 8.7/10
- Languages:** English C1 | German B1 | Bengali native | Hindi fluent